

DSA Practice Tracker (230 Problems)

* marks S.No 1-100 (core set). ** marks priority/must-do problems.

S.No	Topic	Problem	Input	Output	Hint	Status
1 *	Arrays	Find the largest element in an array	[10,25,7,42,18]	42	Traverse once and keep the maximum.	Not Solved
2 *	Arrays	Find the second largest element	[10,25,7,42,18]	25	Track largest and second largest in one pass.	Not Solved
3 *	Arrays	Check if an array is sorted	[1,2,3,4,5]	true	Compare every element with its previous element.	Not Solved
4 *	Arrays	Remove duplicates from a sorted array	[1,1,2,2,3]	[1,2,3]	Use two pointers to overwrite duplicates.	Not Solved
5 *	Arrays	Move all zeroes to the end	[0,1,0,3,12]	[1,3,12,0,0]	Keep a write index for non-zero values.	Not Solved
6 *	Arrays	Left rotate an array by one position	[1,2,3,4,5]	[2,3,4,5,1]	Store the first element and shift left.	Not Solved
7 *	Arrays	Left rotate an array by K positions	arr=[1,2,3,4,5], k=2	[3,4,5,1,2]	Use reversal or k modulo n.	Not Solved
8 *	Arrays	Right rotate an array by K positions	arr=[1,2,3,4,5], k=2	[4,5,1,2,3]	Use reversal or k modulo n.	Not Solved
9 *	Arrays	Find the missing number	[1,2,4,5]	3	Use sum or XOR of indices and values.	Not Solved
10 *	Arrays	Find the number that appears once	[4,1,2,1,2]	4	XOR all values.	Not Solved
11 *	Arrays	Find the maximum and minimum element	[5,2,9,1,7]	Max=9, Min=1	Maintain min and max during one traversal.	Not Solved
12 *	Arrays	Find the second smallest element	[5,2,9,1,7]	2	Track smallest and second smallest.	Not Solved
13 **	Arrays	Find the majority element	[2,2,1,1,1,2,2]	2	Use Boyer-Moore voting.	Not Solved
14 **	Arrays	Find the maximum subarray sum	[-2,1,-3,4,-1,2,1,-5,4]	6	Apply Kadane's algorithm.	Not Solved
15 **	Arrays	Find the maximum profit from buying and selling stock	[7,1,5,3,6,4]	5	Track the minimum price and best profit.	Not Solved
16 *	Arrays	Find the union of two sorted arrays	A=[1,2,4], B=[2,4,6]	[1,2,4,6]	Use two pointers and skip duplicates.	Not Solved
17 *	Arrays	Find the intersection of two arrays	A=[1,2,2,3], B=[2,2,4]	[2,2]	Compare sorted values with two pointers.	Not Solved
18 *	Arrays	Find all leaders in an array	[16,17,4,3,5,2]	[17,5,2]	Traverse from right while tracking the maximum.	Not Solved
19 *	Arrays	Rearrange array elements by sign	[-1,2,-3,4]	[2,4,-1,-3]	Separate positive and negative values, then merge.	Not Solved
20 *	Arrays	Find the maximum consecutive 1s	[1,1,0,1,1,1]	3	Maintain a running count and maximum.	Not Solved
21 *	Arrays	Find the duplicate number	[1,3,4,2,2]	2	Use Floyd's cycle detection or counting.	Not Solved
22 **	Arrays	Find two numbers whose sum equals a target	arr=[2,7,11,15], target=9	[0,1]	Store complements in a HashMap.	Not Solved
23 **	Arrays	Find three numbers whose sum equals zero	[-1,0,1,2,-1,-4]	[[-1,-1,2],[-1,0,1]]	Sort, then fix one value and use two pointers.	Not Solved
24 **	Arrays	Find the longest consecutive sequence	[100,4,200,1,3,2]	4	Use a HashSet and start only at sequence beginnings.	Not Solved

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25 *	Arrays	Find the product of array elements except itself	[1,2,3,4]	[24,12,8,6]	Use prefix and suffix products.	Not Solved
26 *	Strings	Reverse a string	hello	olleh	Use two pointers or StringBuilder.	Not Solved
27 *	Strings	Check whether a string is a palindrome	madam	true	Compare characters from both ends.	Not Solved
28 *	Strings	Count vowels and consonants	Hello World	Vowels=3, Consonants=7	Traverse and classify alphabetic characters.	Not Solved
29 *	Strings	Count frequency of characters	banana	{"a":3,"n":2,"b":1}	Use a HashMap or frequency array.	Not Solved
30 *	Strings	Remove duplicate characters	programming	progamin	Use a boolean set while preserving order.	Not Solved
31 *	Strings	Find the first non-repeating character	swiss	w	Count frequencies, then scan again.	Not Solved
32 *	Strings	Check whether two strings are anagrams	listen,"silent"	true	Compare character frequencies.	Not Solved
33 *	Strings	Find the longest common prefix	flower,"flow","flight"	fl	Shrink the prefix until all strings match.	Not Solved
34 *	Strings	Reverse words in a string	the sky is blue	blue is sky the	Split/scan words and reverse their order.	Not Solved
35 *	Strings	Check whether one string is a rotation of another	waterbottle,"erbottlewat"	true	Check whether s2 occurs in s1+s1.	Not Solved
36 **	Strings	Find the longest substring without repeating characters	abcabcbb	3	Use a sliding window and last-seen positions.	Not Solved
37 *	Strings	Find the longest palindromic substring	babad	bab or "aba"	Expand around every possible center.	Not Solved
38 *	Strings	Count palindromic substrings	aaa	6	Expand around each center and count.	Not Solved
39 *	Strings	Convert Roman numerals to integer	MCMXCIV	1994	Compare each symbol with the next symbol.	Not Solved
40 *	Strings	Convert integer to Roman numeral	(1994)	MCMXCIV	Greedily subtract the largest Roman value.	Not Solved
41 *	Strings	Implement string-to-integer conversion	-42	-42	Skip spaces, read sign, then digits with overflow checks.	Not Solved
42 *	Strings	Check whether two strings are isomorphic	egg,"add"	true	Maintain two-way character mappings.	Not Solved
43 **	Strings	Group anagrams	["eat","tea","tan","ate","nat","bat"]	[["eat","tea","ate"],["tan","nat"],["bat"]]	Use sorted strings or frequency signatures as keys.	Not Solved
44 *	Strings	Find all permutations of a string	abc	["abc","acb","bac","bca","cab","cba"]	Backtrack by choosing unused characters.	Not Solved
45 *	Strings	Implement substring search	hello,"ll"	2	Try matching from each possible starting index.	Not Solved
46 **	HashMap / HashSet	Two Sum	nums=[2,7,11,15], target=9	[0,1]	Store each value's complement in a HashMap.	Not Solved
47 *	HashMap / HashSet	Contains Duplicate	[1,2,3,1]	true	Insert values into a HashSet.	Not Solved
48 *	HashMap / HashSet	Valid Anagram	anagram,"nagaram"	true	Compare frequency maps.	Not Solved
49 *	HashMap / HashSet	Frequency of elements in an array	[1,2,2,3,3,3]	{1:1,2:2,3:3}	Build a HashMap of value to count.	Not Solved
50 *	HashMap / HashSet	First unique character	leetcode	l	Count first, then find the first count of one.	Not Solved

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51 *	HashMap / HashSet	Intersection of two arrays	[1,2,2,1],[2,2]	[2]	Use a set or frequency map.	Not Solved
52 *	HashMap / HashSet	Find duplicate elements	[1,2,3,2,4,3]	[2,3]	Track previously seen values.	Not Solved
53 **	HashMap / HashSet	Longest consecutive sequence	[100,4,200,1,3,2]	4	Use a HashSet and extend only from sequence starts.	Not Solved
54 **	HashMap / HashSet	Group Anagrams	["eat","tea","tan","ate","nat","bat"]	grouped anagrams	Identify the pattern, choose the appropriate data structure, and handle edge cases.	Not Solved
55 **	HashMap / HashSet	Subarray Sum Equals K	nums=[1,2,3], k=3	2	Store prefix-sum frequencies.	Not Solved
56 *	HashMap / HashSet	Longest subarray with sum zero	nums=[1,-1,3,1,-4],	4	Store the first index for each prefix sum.	Not Solved
57 *	HashMap / HashSet	Count number of subarrays with given sum	nums=[1,1,1], k=2	2	Use prefix sum plus frequency map.	Not Solved
58 *	HashMap / HashSet	Top K Frequent Elements	[1,1,1,2,2,3], k=2	[1,2]	Count frequencies and use a heap/buckets.	Not Solved
59 *	HashMap / HashSet	Isomorphic Strings	egg,"add"	true	Maintain mappings in both directions.	Not Solved
60 *	HashMap / HashSet	Longest substring with at most K distinct characters	eceba, k=2	3	Expand and shrink a frequency-map window.	Not Solved
61 *	Two Pointers	Two Sum in a sorted array	nums=[2,7,11,15], target=9	[0,1]	Move left/right pointers based on the sum.	Not Solved
62 *	Two Pointers	Remove duplicates from sorted array	[1,1,2,2,3]	[1,2,3]	Use slow and fast pointers.	Not Solved
63 *	Two Pointers	Move zeroes	[0,1,0,3,12]	[1,3,12,0,0]	Write non-zero values first, then fill zeros.	Not Solved
64 *	Two Pointers	Valid palindrome	A man, a plan, a canal: Panama	true	Use two pointers while ignoring non-alphanumeric characters.	Not Solved
65 *	Two Pointers	Reverse vowels of a string	hello	eo	Move two pointers until both point to vowels.	Not Solved
66 **	Two Pointers	Container With Most Water	[1,8,6,2,5,4,8,3,7]	49	Move the pointer at the shorter height.	Not Solved
67 **	Two Pointers	3Sum	[-1,0,1,2,-1,-4]	[[-1,-1,2],[-1,0,1]]	Sort, fix one number, and use two pointers.	Not Solved
68 *	Two Pointers	4Sum	[1,0,-1,0,-2,2], target=0	[[-2,-1,1,2],[-2,0,0,2],[-1,0,0,1]]	Sort, fix two numbers, and use two pointers.	Not Solved
69 *	Two Pointers	Sort colors	[2,0,2,1,1,0]	[0,0,1,1,2,2]	Use the Dutch National Flag algorithm.	Not Solved
70 **	Two Pointers	Trapping Rain Water	[4,2,0,3,2,5]	9	Use two pointers with left/right maximums.	Not Solved
71 *	Sliding Window	Maximum sum subarray of size K	[2,1,5,1,3,2], k=3	9	Maintain a fixed-size sliding window sum.	Not Solved
72 *	Sliding Window	Maximum number of consecutive 1s	[1,1,1,0,0,1,1]	3	Use a running count of ones.	Not Solved
73 **	Sliding Window	Longest substring without repeating characters	abcabcbb	3	Identify the pattern, choose the appropriate data structure, and handle edge cases.	Not Solved
74 *	Sliding Window	Longest substring with at most K distinct characters	eceba, k=2	3	Expand and shrink a frequency-map window.	Not Solved

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75 *	Sliding Window	Minimum size subarray sum	[2,3,1,2,4,3], target=7	2	Shrink the window while sum meets the target.	Not Solved
76 *	Sliding Window	Maximum average subarray	[1,12,-5,-6,50,3], k=4	12.75	Maximize the fixed-size window sum.	Not Solved
77 *	Sliding Window	Permutation in String	ab,"eidbaooo"	true	Compare fixed-size frequency windows.	Not Solved
78 *	Sliding Window	Find all anagrams in a string	cbaebabacd,"abc"	[0,6]	Use a sliding frequency window.	Not Solved
79 **	Sliding Window	Minimum Window Substring	ADOBECODEBANC,"ABC"	BANC	Expand until valid, then shrink for minimum length.	Not Solved
80 *	Sliding Window	Longest Repeating Character Replacement	AABABBA, k=1	4	Track max frequency in the current window.	Not Solved
81 **	Binary Search	Binary Search	nums=[-1,0,3,5,9,12], target=9	4	Compare the target with the middle element.	Not Solved
82 *	Binary Search	Search Insert Position	nums=[1,3,5,6], target=2	1	Binary search for the first position >= target.	Not Solved
83 *	Binary Search	Find first and last position of an element	nums=[5,7,7,8,8,10], target=8	[3,4]	Run binary search twice for bounds.	Not Solved
84 *	Binary Search	Count occurrences in a sorted array	nums=[1,2,2,2,3], target=2	3	Find first and last positions.	Not Solved
85 *	Binary Search	Find the first occurrence	nums=[1,2,2,3], target=2	1	When matched, continue searching left.	Not Solved
86 *	Binary Search	Find the last occurrence	nums=[1,2,2,3], target=2	2	When matched, continue searching right.	Not Solved
87 **	Binary Search	Search in a rotated sorted array	nums=[4,5,6,7,0,1,2], target=0	4	Identify which half is sorted.	Not Solved
88 *	Binary Search	Find minimum in a rotated sorted array	[4,5,6,7,0,1,2]	0	Use the sorted-half property.	Not Solved
89 *	Binary Search	Find the rotation count	[15,18,2,3,6,12]	2	The minimum element's index is the rotation count.	Not Solved
90 *	Binary Search	Find a peak element	[1,2,1,3,5,6,4]	5	Move toward the side with a larger neighbor.	Not Solved
91 *	Binary Search	Find square root of a number	16	4	Binary search the integer answer.	Not Solved
92 *	Binary Search	Find Kth missing positive number	arr=[2,3,4,7,11], k=5	9	Binary search using missing count.	Not Solved
93 **	Binary Search	Koko Eating Bananas	piles=[3,6,7,11], h=8	4	Binary search the minimum eating speed.	Not Solved
94 *	Binary Search	Capacity to Ship Packages Within D Days	weights=[1,2,3,4,5,6,7,8,9,10], days=5	15	Binary search the minimum capacity.	Not Solved
95 *	Binary Search	Median of Two Sorted Arrays	A=[1,3], B=[2]	2.0	Binary search the partition in the smaller array.	Not Solved
96 *	Linked List	Create and traverse a linked list	[1,2,3]	1 -> 2 -> 3	Create nodes and follow next pointers.	Not Solved
97 *	Linked List	Insert a node at beginning	1 -> 2 -> 3	0 -> 1 -> 2 -> 3	Point the new node to the current head.	Not Solved
98 *	Linked List	Insert a node at end	1 -> 2 -> 3	1 -> 2 -> 3 -> 4	Traverse to the last node and attach the new node.	Not Solved
99 *	Linked List	Delete a node	1 -> 2 -> 3, delete 2	1 -> 3	Reconnect the previous node to the next node.	Not Solved
100 *	Linked List	Search for an element	1 -> 2 -> 3, search 2	true	Traverse until the value is found.	Not Solved
101	Linked List	Reverse a linked list	1 -> 2 -> 3	3 -> 2 -> 1	Reverse next pointers iteratively.	Not Solved
102	Linked List	Find the middle of a linked list	1 -> 2 -> 3 -> 4 -> 5	3	Use slow and fast pointers.	Not Solved
103	Linked List	Detect a cycle	1 -> 2 -> 3 -> 4, cycle at 2	true	Use Floyd's slow and fast pointers.	Not Solved

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104	Linked List	Find the starting point of a cycle	1 -> 2 -> 3 -> 4, cycle at 2	node 2	After detection, reset one pointer to head.	Not Solved
105	Linked List	Merge two sorted linked lists	1 -> 2 -> 4, 1 -> 3 -> 4	1 -> 1 -> 2 -> 3 -> 4 -> 4	Choose the smaller current node each time.	Not Solved
106	Linked List	Remove Nth node from the end	1 -> 2 -> 3 -> 4 -> 5, n=2	1 -> 2 -> 3 -> 5	Use two pointers with an n-node gap.	Not Solved
107	Linked List	Check whether linked list is palindrome	1 -> 2 -> 2 -> 1	true	Find middle, reverse second half, compare.	Not Solved
108	Linked List	Add two numbers represented by linked lists	2 -> 4 -> 3 + 5 -> 6 -> 4	7 -> 0 -> 8	Add digit by digit with carry.	Not Solved
109	Linked List	Sort a linked list	4 -> 2 -> 1 -> 3	1 -> 2 -> 3 -> 4	Use merge sort for linked lists.	Not Solved
110	Linked List	Find intersection of two linked lists	1 -> 2 -> 3 -> 4, 9 -> 3 -> 4	node 3	Switch heads when pointers reach the end.	Not Solved
111	Stack & Queue	Implement stack using array	[1,2,3]	push/pop gives 3,2,1	Maintain a top index.	Not Solved
112	Stack & Queue	Implement queue using array	[1,2,3]	dequeue gives 1,2,3	Maintain front/rear indices or circular storage.	Not Solved
113	Stack & Queue	Implement stack using linked list	[1,2,3]	push/pop gives 3,2,1	Push and pop at the head.	Not Solved
114	Stack & Queue	Implement queue using linked list	[1,2,3]	dequeue gives 1,2,3	Maintain front and rear pointers.	Not Solved
115	Stack & Queue	Implement stack using queues	push 1,2,3 then pop	3,2,1	Use one or two queues to simulate LIFO.	Not Solved
116	Stack & Queue	Implement queue using stacks	enqueue 1,2,3 then dequeue	1,2,3	Use input and output stacks.	Not Solved
117	Stack & Queue	Valid Parentheses	(){}[]	true	Push opening brackets and match closing brackets.	Not Solved
118	Stack & Queue	Min Stack	push 5,3,7; getMin	3	Store current minimum along with stack values.	Not Solved
119	Stack & Queue	Next Greater Element	[4,5,2,25]	[5,25,25,-1]	Use a monotonic decreasing stack.	Not Solved
120	Stack & Queue	Next Smaller Element	[4,5,2,25]	[2,2,-1,-1]	Use a monotonic increasing stack.	Not Solved
121	Stack & Queue	Stock Span Problem	[100,80,60,70,60,75,85]	[1,1,1,2,1,4,6]	Maintain a stack of indices with decreasing prices.	Not Solved
122	Stack & Queue	Daily Temperatures	[73,74,75,71,69,72,76,73]	[1,1,4,2,1,1,0,0]	Use a monotonic stack of unresolved indices.	Not Solved
123	Stack & Queue	Largest Rectangle in Histogram	[2,1,5,6,2,3]	10	Use a monotonic stack of bar indices.	Not Solved
124	Stack & Queue	Evaluate Reverse Polish Notation	["2","1","+","3","*"]	9	Push operands and apply operators.	Not Solved
125	Stack & Queue	Sliding Window Maximum	[1,3,-1,-3,5,6,7], k=3	[3,3,5,5,6,7]	Use a decreasing deque of indices.	Not Solved
126	Recursion & Backtracking	Print numbers using recursion	n=5	1 2 3 4 5	Print n after recursively processing n-1.	Not Solved
127	Recursion & Backtracking	Calculate factorial	n=5	120	Base case n<=1; multiply by factorial(n-1).	Not Solved
128	Recursion & Backtracking	Calculate Fibonacci	n=7	13	Use recursion for practice; later optimize with memoization.	Not Solved
129	Recursion & Backtracking	Reverse a string recursively	hello	olleh	Swap ends and recurse inward.	Not Solved
130	Recursion & Backtracking	Check palindrome recursively	madam	true	Compare ends and recurse inward.	Not Solved

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131	Recursion & Backtracking	Sum of digits using recursion	12345	15	Add last digit to the recursive result.	Not Solved
132	Recursion & Backtracking	Calculate power recursively	2^10	1024	Use exponentiation by squaring.	Not Solved
133	Recursion & Backtracking	Generate all subsequences	[1,2,3]	[],[1],[2],[3],[1,2],[1,3],[2,3],[1,2,3]]	At each index choose include or exclude.	Not Solved
134	Recursion & Backtracking	Generate all subsets	[1,2]	[],[1],[2],[1,2]]	Backtrack with include/exclude choices.	Not Solved
135	Recursion & Backtracking	Generate all permutations	[1,2,3]	6 permutations	Swap each candidate into the current position.	Not Solved
136	Recursion & Backtracking	Combination Sum	candidates=[2,3,6,7], target=7	[[2,2,3],[7]]	Backtrack while remaining target is non-negative.	Not Solved
137	Recursion & Backtracking	Letter combinations of a phone number	23	['ad','ae','af','bd',...]	Map each digit to letters and backtrack.	Not Solved
138	Recursion & Backtracking	Generate valid parentheses	n=3	((())),"(())",...	Track open/close counts and only add valid choices.	Not Solved
139	Recursion & Backtracking	N-Queens	n=4	2 solutions	Place one queen per row and check columns/diagonals.	Not Solved
140	Recursion & Backtracking	Sudoku Solver	4x4 Sudoku board	Solved valid Sudoku	Try valid digits and backtrack on conflicts.	Not Solved
141	Trees / BST	Create a binary tree	[1,2,3]	root with children 2 and 3	Build nodes and connect left/right children.	Not Solved
142	Trees / BST	Preorder traversal	[1,2,3]	[1,2,3]	Visit root, left, then right.	Not Solved
143	Trees / BST	Inorder traversal	[1,2,3]	[2,1,3]	Visit left, root, then right.	Not Solved
144	Trees / BST	Postorder traversal	[1,2,3]	[2,3,1]	Visit left, right, then root.	Not Solved
145	Trees / BST	Level-order traversal	[1,2,3,4,5]	[[1],[2,3],[4,5]]	Use a queue level by level.	Not Solved
146	Trees / BST	Maximum depth of binary tree	[3,9,20,null,null,15,7]	3	Depth = 1 + max(left depth, right depth).	Not Solved
147	Trees / BST	Minimum depth of binary tree	[3,9,20,null,null,15,7]	2	Find the shortest root-to-leaf path.	Not Solved
148	Trees / BST	Count nodes in a binary tree	[1,2,3]	3	Return 1 plus counts of both subtrees.	Not Solved
149	Trees / BST	Count leaf nodes	[1,2,3]	2	Count nodes with no children.	Not Solved
150	Trees / BST	Check if two trees are identical	[1,2,3] vs [1,2,3]	true	Compare values and both subtrees recursively.	Not Solved
151	Trees / BST	Invert a binary tree	[4,2,7,1,3,6,9]	[4,7,2,9,6,3,1]	Swap left and right recursively.	Not Solved
152	Trees / BST	Check whether tree is balanced	[3,9,20,null,null,15,7]	true	Return height and detect imbalance bottom-up.	Not Solved
153	Trees / BST	Diameter of binary tree	[1,2,3,4,5]	3	Track max left height + right height.	Not Solved
154	Trees / BST	Maximum path sum	[-10,9,20,null,null,15,7]	42	Compute best downward path and update global maximum.	Not Solved
155	Trees / BST	Zigzag level-order traversal	[3,9,20,null,null,15,7]	[[3],[20,9,15,7]]	Alternate traversal direction at each level.	Not Solved

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156	Trees / BST	Right view of binary tree	[1,2,3,null,5,null,4]	[1,3,4]	Take the last visible node at each level.	Not Solved
157	Trees / BST	Left view of binary tree	[1,2,3,null,5,null,4]	[1,3,4]	Take the first visible node at each level.	Not Solved
158	Trees / BST	Boundary traversal	[1,2,3,4,5,null,6,7]	boundary order	Collect root, left boundary, leaves, and reversed right boundary.	Not Solved
159	Trees / BST	Vertical order traversal	[3,9,20,null,null,15,7]	vertical columns	Group nodes by horizontal distance.	Not Solved
160	Trees / BST	Lowest Common Ancestor	root=[3,5,1,6,2,0,8,null,null,7,4], p=5,q=1	3	Return when either target is found and combine subtree results.	Not Solved
161	Trees / BST	Validate Binary Search Tree	[2,1,3]	true	Carry valid lower and upper bounds.	Not Solved
162	Trees / BST	Search in BST	[4,2,7,1,3] target=3	3	Choose left or right based on target.	Not Solved
163	Trees / BST	Insert into BST	[4,2,7,1,3] insert 5	[4,2,7,1,3,5]	Recursively find the correct null position.	Not Solved
164	Trees / BST	Delete node from BST	[5,3,6,2,4] delete 3	BST after deletion	Handle zero/one child and replace two-child node with successor.	Not Solved
165	Trees / BST	Kth smallest element in BST	[5,3,6,2,4,1] k=3	4	Inorder traversal visits BST nodes in sorted order.	Not Solved
166	Heap / Priority Queue	Implement a Min Heap	[3,1,6,5,2,4]	[1,2,4,5,3,6] min-heap	Maintain heap order while bubbling up/down.	Not Solved
167	Heap / Priority Queue	Implement a Max Heap	[3,1,6,5,2,4]	[6,5,4,1,2,3] max-heap	Maintain reverse heap order while bubbling up/down.	Not Solved
168	Heap / Priority Queue	Find Kth largest element	[3,2,1,5,6,4], k=2	5	Use a min-heap of size k.	Not Solved
169	Heap / Priority Queue	Find Kth smallest element	[3,2,1,5,6,4], k=2	2	Use a max-heap of size k.	Not Solved
170	Heap / Priority Queue	Top K frequent elements	[1,1,1,2,2,3], k=2	[1,2]	Count and keep the k highest frequencies.	Not Solved
171	Heap / Priority Queue	Merge K sorted arrays	[[1,4,5],[1,3,4],[2,6]]	[1,1,2,3,4,4,5,6]	Use a min-heap containing the current element from each array.	Not Solved
172	Heap / Priority Queue	Merge K sorted linked lists	[1,4,5],[1,3,4],[2,6]	[1,1,2,3,4,4,5,6]	Put each list head into a min-heap.	Not Solved
173	Heap / Priority Queue	Find median from a data stream	[1,2,3] then add 4	2.5	Use two heaps: max-heap left and min-heap right.	Not Solved
174	Heap / Priority Queue	K closest points to origin	[[1,3],[2,2],[3,1]], k=2	[[1,3],[2,2]]	Compare squared distances and maintain k points.	Not Solved
175	Heap / Priority Queue	Task Scheduler	tasks=[A,A,A,B,B,B], n=2	8	Use frequency counts and a max-heap.	Not Solved
176	Graphs	Represent a graph using adjacency matrix	V=3, edges=(0,1),(1,2)	3x3 adjacency matrix	Create VxV and mark edges.	Not Solved
177	Graphs	Represent a graph using adjacency list	V=3, edges=(0,1),(1,2)	0:[1],1:[0,2],2:[1]	Create a list of neighbors for every vertex.	Not Solved
178	Graphs	Breadth First Search — BFS	V=5, edges=(0,1),(0,2),(1,3),(2,4), start=0	[0,1,2,3,4]	Use a queue and visited array.	Not Solved
179	Graphs	Depth First Search — DFS	same graph, start=0	[0,1,3,2,4]	Use recursion or an explicit stack.	Not Solved

S.No	Topic	Problem	Input	Output	Hint	Status
180	Graphs	Number of connected components	V=5, edges=(0,1),(1,2),(3,4)	2 components	Run DFS/BFS from each unvisited vertex.	Not Solved
181	Graphs	Number of islands	grid=[[1,1,0],[0,1,0],[1,0,1]]	3	DFS/BFS each unvisited land cell.	Not Solved
182	Graphs	Flood Fill	image=[[1,1,1],[1,1,0],[1,0,1]], sr=1,sc=1,color=2	[[2,2,2],[2,2,0],[2,0,1]]	DFS/BFS from the starting cell and recolor matching cells.	Not Solved
183	Graphs	Detect cycle in an undirected graph	edges=(0,1),(1,2),(2,0)	true	Track parent during DFS/BFS.	Not Solved
184	Graphs	Detect cycle in a directed graph	edges=0->1,1->2,2->0	true	Use a recursion-stack/3-state array.	Not Solved
185	Graphs	Topological Sort using DFS	0->1,0->2,1->3,2->3	[0,2,1,3]	Push a node after visiting all dependencies.	Not Solved
186	Graphs	Topological Sort using BFS	same DAG	[0,2,1,3]	Use indegrees and Kahn's algorithm.	Not Solved
187	Graphs	Course Schedule	courses=2, prereq=[[1,0]]	true	Check whether prerequisite graph has a cycle.	Not Solved
188	Graphs	Shortest path in an unweighted graph	edges 0-1,1-2, start=0	[0,1,2]	Use BFS from the source.	Not Solved
189	Graphs	Dijkstra's Algorithm	graph 0->1(4),0->2(1),2->1(2)	distance to 1 = 3	Relax edges using a min-priority queue.	Not Solved
190	Graphs	Bellman-Ford Algorithm	V=3, edges 0->1=4,0->2=5	shortest distances	Relax all edges V-1 times and check one more time.	Not Solved
191	Graphs	Floyd-Warshall Algorithm	graph with negative edge	shortest distances / detect negative cycle	Try every vertex as an intermediate node.	Not Solved
192	Graphs	Minimum Spanning Tree — Prim's Algorithm	graph 3x3	all-pairs shortest path matrix	Grow the MST using the cheapest outgoing edge.	Not Solved
193	Graphs	Minimum Spanning Tree — Kruskal's Algorithm	weighted graph	MST total weight	Sort edges and use Union-Find.	Not Solved
194	Graphs	Word Ladder	weighted graph	MST total weight	BFS through one-letter transformations.	Not Solved
195	Graphs	Rotten Oranges	hit,"hot","dot","dog","cog"	5	Multi-source BFS from all rotten oranges.	Not Solved
196	Greedy	Assign Cookies	grid=[[2,1,1],[1,1,0],[0,1,1]]	4 minutes	Sort both arrays and greedily match smallest suitable cookie.	Not Solved
197	Greedy	Lemonade Change	g=[1,2,3], s=[1,1]	1	Track available \$5 and \$10 bills.	Not Solved
198	Greedy	Jump Game	[5,5,5,10,20]	true	Track the farthest reachable index.	Not Solved
199	Greedy	Jump Game II	[2,3,1,1,4]	true	Expand the current reachable range greedily.	Not Solved
200	Greedy	Gas Station	[2,3,1,1,4]	2 jumps	Reset start when cumulative gas becomes negative.	Not Solved
201	Greedy	Activity Selection	gas=[1,2,3,4,5], cost=[3,4,5,1,2]	3	Sort by finish time and select compatible activities.	Not Solved
202	Greedy	Fractional Knapsack	activities=(1,2),(3,4),(0,6),(5,7),(8,9)	3 activities	Sort by value/weight ratio.	Not Solved
203	Greedy	Job Sequencing Problem	values=[60,100,120], weights=[10,20,30], W=50	240	Schedule high-profit jobs in latest available slots.	Not Solved
204	Greedy	Minimum number of coins	jobs with deadlines/profits	maximum profit schedule	Greedily take largest coins when the denomination system allows it.	Not Solved
205	Greedy	Merge Overlapping Intervals	coins=[1,2,5], amount=11	3	Sort by start and merge overlapping ranges.	Not Solved

S.No	Topic	Problem	Input	Output	Hint	Status
206	Dynamic Programming	Fibonacci using DP	[[1,3],[2,6],[8,10],[15,18]]	[[1,6],[8,10],[15,18]]	Store previously computed Fibonacci values.	Not Solved
207	Dynamic Programming	Climbing Stairs	n=7	13	dp[i] = dp[i-1] + dp[i-2].	Not Solved
208	Dynamic Programming	Frog Jump	n=5	8	Try jumps of allowed sizes and memoize states.	Not Solved
209	Dynamic Programming	Maximum Sum of Non-Adjacent Elements	heights=[10,20,30,10]	20	Choose current plus dp[i-2] or skip current.	Not Solved
210	Dynamic Programming	House Robber	[2,7,9,3,1]	12	Use the same non-adjacent maximum recurrence.	Not Solved
211	Dynamic Programming	House Robber II	[2,7,9,3,1]	12	Solve two linear cases: exclude first or exclude last.	Not Solved
212	Dynamic Programming	Best Time to Buy and Sell Stock	[2,3,2]	3	Track minimum price and maximum profit.	Not Solved
213	Dynamic Programming	Coin Change	prices=[7,1,5,3,6,4]	5	dp[amount] = minimum coins over all denominations.	Not Solved
214	Dynamic Programming	Coin Change II	coins=[1,2,5], amount=11	3	Count combinations using coin-based DP.	Not Solved
215	Dynamic Programming	0/1 Knapsack	coins=[1,2,5], amount=5	4	For each item choose take or skip.	Not Solved
216	Dynamic Programming	Unbounded Knapsack	weights=[1,2,3], values=[6,10,12], W=5	22	Allow taking the same item multiple times.	Not Solved
217	Dynamic Programming	Subset Sum	weights=[2,3,4], values=[40,50,60], W=6	120	Track whether each sum is reachable.	Not Solved
218	Dynamic Programming	Partition Equal Subset Sum	nums=[2,3,7,8,10], target=11	true	Reduce to subset sum of total/2.	Not Solved
219	Dynamic Programming	Count subsets with given sum	[1,5,11,5]	true	DP counts ways to form each sum.	Not Solved
220	Dynamic Programming	Longest Common Subsequence	nums=[1,2,3], target=3	2 subsets	Match characters or take the better previous state.	Not Solved
221	Dynamic Programming	Longest Common Substring	abcde,"ace"	3	Reset length to zero when characters differ.	Not Solved
222	Dynamic Programming	Longest Increasing Subsequence	abc,"ab"	2	Use DP or patience sorting.	Not Solved
223	Dynamic Programming	Longest Palindromic Subsequence	[10,9,2,5,3,7,101,18]	4	Use interval DP on matching endpoints.	Not Solved
224	Dynamic Programming	Edit Distance	bbbab	4	Use insert, delete, and replace transitions.	Not Solved
225	Dynamic Programming	Minimum Path Sum	horse,"ros"	3	Take minimum cost from top or left.	Not Solved

S.No	Topic	Problem	Input	Output	Hint	Status
226	Dynamic Programming	Unique Paths	grid=[[1,3,1],[1,5,1],[4,2,1]]	7	$dp[i][j] = dp[i-1][j] + dp[i][j-1]$.	Not Solved
227	Dynamic Programming	Unique Paths II	m=3,n=7	28	Same recurrence, but blocked cells have zero ways.	Not Solved
228	Dynamic Programming	Triangle	grid=[[0,0,0],[0,1,0],[0,0,0]]	2	Bottom-up DP gives the minimum path.	Not Solved
229	Dynamic Programming	Matrix Chain Multiplication	[[2],[3,4],[6,5,7],[4,1,8,3]]	11	Try every split point and minimize multiplication cost.	Not Solved
230	Dynamic Programming	Word Break	dims=[10,20,30,40,30]	26000	dp[i] indicates whether prefix i can be segmented.	Not Solved